

Lecture 9/17 Notes

• Review: Scoring scheme -

S	-	A	C	G	T
-	0	0	0	0	0
A	0	1	0	0	0
C	0	0	1	0	0
G	0	0	0	1	0
T	0	0	0	0	1

Unit scoring

↳ Basic mechanism of scoring alignment

• Review: Edit graph - $G_{x,y}$ - an edge labelled directed graph used to find optimal alignment

↳ $G = (V, E)$ where V = vertex set and E = edge set

↳ $(M+1) \times (N+1)$

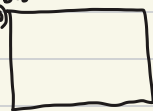
↳ gap in Y , gap in X , alignment

- Conclusion: Only 3 types of edges coming to (i, j)

- Theorem: There is a 1:1 correspondence between directed path from $(0,0)$ to (M,N) and alignment between X and Y

- Edit graphs are Directed Acyclic Graph (DAG)

For (i,j)



• Def - Denote $S(i,j)$ the score of max path from (Begin) to (i,j)

↳ Here are only 3 edges directed into vertex (i,j) , any optimal path P_{opt} to (i,j) must use one of these edges, from $(i-1,j)$, $(i,j-1)$, or $(i-1,j-1)$

$$S(i,j) = \max \begin{cases} S(i-1,j-1) + S(x_i, y_j) \\ S(i-1,j) + S(x_i, -) \\ S(i,j-1) + S(-, y_j) \end{cases}$$

• Need to calculate $S(i,j)$ such that we have already computed the values of S needed by the recurrence
- Can be done column by column

• Algorithm for Global Alignment

$$S[0,0] = 0$$

for $j = 1$ to N

$$S[0,j] = S[0,j-1] + S(-, y_j)$$

for $i = 1$ to M

$$S[i,0] = S[i-1,0] + S(x_i, -)$$

$$\text{for } j = 1 \text{ to } N \quad \left[\begin{array}{l} S[i,j] = \max \begin{cases} S[i-1,j-1] + S(x_i, y_j) \\ S[i-1,j] + S(x_i, -) \\ S[i,j-1] + S(-, y_j) \end{cases} \end{array} \right]$$

Lecture 9/17 cont.

• Alignment of multiple proteins (multiple alignment)

- Ex: $\left. \begin{array}{l} \text{BABA} \\ \text{AAAC} \\ \text{AAEC} \\ \text{AABA} \\ \text{AACC} \\ \text{AABC} \end{array} \right\} \begin{array}{l} 14 \text{ As} = \frac{14}{24} \text{ A} \\ 4 \text{ Bs} = \frac{4}{24} \text{ B observed} \\ 6 \text{ Cs} = \frac{6}{24} \text{ C} \end{array}$ # of aligned pairs = $4 \times \binom{6}{2} = 60$

Aligned Pair	Frequency Observed
A to A	$\frac{26}{60}$
A to B	$\frac{8}{60}$
A to C	$\frac{10}{60}$
B to B	$\frac{3}{60}$
B to C	$\frac{6}{60}$
C to C	$\frac{7}{60}$

• Compare observed frequency to expected frequency

↳ Generate random sequences using observed frequency and make an alignment

Expected proportions:

A to A: $\frac{14}{24} \times \frac{14}{24}$

A to B: $2 \times \frac{14}{24} \times \frac{4}{24}$

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Used to calculate estimated likelihood ratio

$$2 \cdot \log_2 \left(\frac{\text{Observed}}{\text{Expected}} \right)$$

• New scoring scheme

A B C
A 1 -1 -2
B -1 2 1
C -2 1 2

Aligned Pair	Observed Freq	Expected Freq	Estimated likelihood ratio
A to A	$\frac{26}{60}$	$\frac{196}{576}$	0.20
A to B	$\frac{8}{60}$	$\frac{112}{576}$	-1.09
A to C	$\frac{10}{60}$	$\frac{168}{576}$	-1.61
B to B	$\frac{3}{60}$	$\frac{16}{576}$	1.20
B to C	$\frac{6}{60}$	$\frac{48}{576}$	0.53
C to C	$\frac{7}{60}$	$\frac{46}{576}$	1.83